

William Phong

williamphong10@gmail.com | linkedin.com/in/williamphong | waphong.com | github.com/williamphong

EXPERIENCE

Machine Learning Intern

June 2026 – Present

Sesh Incorporated

San Diego, CA

- Built a gradient-boosted decision tree model as a benchmark against internal models for immuno-oncology
- Preprocessed and trained a pipeline over patient and alignment data in text and WIG formats, then analyzed model performance against internal baselines

Graduate Research Assistant & Project Manager

Sep 2025 – Present

SDSU Research Foundation — Climate Informatics Lab (SCIL)

San Diego, CA

- Founding engineer on iCHARM, a full-stack platform for 3D visualization and statistical analysis of climate data; drove it from prototype to production under Prof. Samuel Shen
- Designed and built a data pipeline serving 6+ TB of NOAA climate datasets on self-hosted SDSU infrastructure using Zarr/NetCDF/Parquet, PostgreSQL, FastAPI, and Docker
- Built a responsive Next.js/TypeScript frontend that renders climate rasters on an interactive 3D globe
- Shipped five in-browser analysis tools — time series, trend, histogram, periodogram, and spectrogram
- Engineered a multi-tenant backend with per-site PostgreSQL data isolation
- Led 3 engineering teams through release while owning architecture and code review

Data Science Intern

Jun 2025 – Aug 2025

Tensor Therapeutics

San Diego, CA

- Built drug discovery workflows in Python with Boltz-2 to predict RNA structures and molecular interactions
- Deployed a variant calling pipeline using nf-core/Sarek and Docker for genomic data analysis
- Managed GPU virtual machines on Azure and Lambda.ai, including secure networking

PROJECTS

PromptCARLA | *Next.js, TypeScript, Python, Docker*

- Built an agentic prompt-to-config interface that turns natural language into validated CARLA traffic simulation scenarios; first author on a paper accepted to IMNS'26 (IEEE proceedings)
- Benchmarked four LLM backends across a 59-prompt, three-tier evaluation suite, reaching 98.31% schema validation success
- Designed a multi-service architecture (agent, simulator, snapshot) orchestrated with Docker Compose

CSUSM Student Portal Application | *Java, Android Studio, MySQL*

- Interviewed CSUSM faculty, admin, and students to determine customer requirements and feature scope
- Delivered the application with a MySQL database and hashed credential storage, running agile scheduling and authoring the SRS

Spotify Day List Word Cloud | *Python, AWS RDS, Matplotlib*

- Built a tool that generates word clouds over time from Spotify day list titles
- Implemented the OAuth 2.0 authorization code flow against the Spotify API and persisted JSON responses in AWS RDS PostgreSQL

EDUCATION

San Diego State University

Aug 2025 – May 2027

Master of Science in Computer Science

GPA: 3.63

California State University San Marcos

May 2024

Bachelor of Science in Computer Science

GPA: 3.87

- **Honors:** Cum Laude, Dean's List Spring 2023 & 2024

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, C++, C, C#, Java

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Tailwind CSS, Cesium.js, Pandas, Matplotlib

Infrastructure & Tools: Docker, Git, PostgreSQL, Linux, Nginx, Zarr/NetCDF, AWS, Azure, Google Cloud, Cloudflare